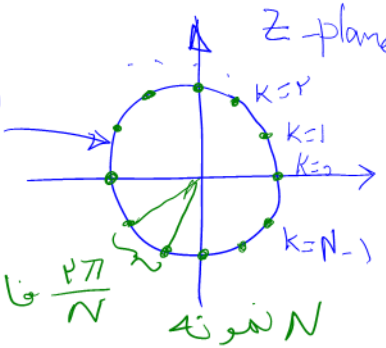
$x_c(t)$ نمونه برداری $\longrightarrow$ $x[n] \longrightarrow X(e^{j\omega}) = \sum x[n]\, e^{-j\omega n}$ ω کمیت پیوسته

Discrete-time FT $\longrightarrow$ DTFT

THE DISCRETE FOURIER SERIES



$\text{DTFT}\Big|_{\omega=\omega_k = k\frac{2\pi}{N}} = \text{DFS}$

$k = 0, 1, \dots, N-1$

$$\boxed{W_N = e^{-j(2\pi/N)}.}$$

the DFS analysis–synthesis pair is expressed as

حوزه زمان ← فرکانس

$$\text{Analysis equation:} \quad \tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn}.$$

Kernel

کرنل تبدیل

متغیر فرکانس گسسته

$$\text{Synthesis equation:} \quad \tilde{x}[n] = \boxed{\frac{1}{N}} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn}.$$

مقیاس

متناوب با دوره تناوب N

TABLE 8.1 SUMMARY OF PROPERTIES OF THE DFS

	Periodic Sequence (Period N)	DFS Coefficients (Period N)
1.	$\tilde{x}[n]$ متناوب با دوره تناوب N	$\tilde{X}[k]$ periodic with period N
2.	$\tilde{x}_1[n], \tilde{x}_2[n]$	$\tilde{X}_1[k], \tilde{X}_2[k]$ periodic with period N
3.	خطی $a\tilde{x}_1[n] + b\tilde{x}_2[n]$	$a\tilde{X}_1[k] + b\tilde{X}_2[k]$
4.	دوبار سری فوریه $\tilde{X}[n]$	$N\tilde{x}[-k]$
5.	جابجایی در زمان $\tilde{x}[n-m]$	$W_N^{km}\tilde{X}[k]$
6.	$W_N^{-\ell n}\tilde{x}[n]$	$\tilde{X}[k-\ell]$ جابجایی در فرکانس
7.	کانولوشن در زمان $\sum_{m=0}^{N-1}\tilde{x}_1[m]\tilde{x}_2[n-m]$ (periodic convolution) Circular Convolution	$\tilde{X}_1[k]\tilde{X}_2[k]$ ضرب
8.	دوگان $\tilde{x}_1[n]\tilde{x}_2[n]$	$\frac{1}{N}\sum_{\ell=0}^{N-1}\tilde{X}_1[\ell]\tilde{X}_2[k-\ell]$ (periodic convolution)
9.	$\tilde{x}^*[n]$ مزدوج مختلط	$\tilde{X}^*[-k]$ مزدوج انعکاس یافته

$$\left(e^{-j\frac{2\pi}{N}}\right)^* = e^{j\frac{2\pi}{N}}$$

TABLE 8.1 *(Continued)*

	Periodic Sequence (Period N)	DFS Coefficients (Period N)
10.	$\tilde{x}^*[-n]$	مزدوج در فرکانس $\tilde{X}^*[k]$
11.	$\mathcal{R}e\{\tilde{x}[n]\}$ بخش حقیقی	بخش زوج سری فوریه $\tilde{X}_e[k] = \frac{1}{2}(\tilde{X}[k] + \tilde{X}^*[-k])$
12.	$j\mathcal{J}m\{\tilde{x}[n]\}$ // موهومی	// // فرد // // $\tilde{X}_o[k] = \frac{1}{2}(\tilde{X}[k] - \tilde{X}^*[-k])$
13.	$\tilde{x}_e[n] = \frac{1}{2}(\tilde{x}[n] + \tilde{x}^*[-n])$	$\mathcal{R}e\{\tilde{X}[k]\}$
14.	$\tilde{x}_o[n] = \frac{1}{2}(\tilde{x}[n] - \tilde{x}^*[-n])$	$j\mathcal{J}m\{\tilde{X}[k]\}$
	Properties 15–17 apply only when $x[n]$ is real.	
15.	Symmetry properties for $\tilde{x}[n]$ real.	$\begin{cases} \tilde{X}[k] = \tilde{X}^*[-k] \\ \mathcal{R}e\{\tilde{X}[k]\} = \mathcal{R}e\{\tilde{X}[-k]\} \\ \mathcal{J}m\{\tilde{X}[k]\} = -\mathcal{J}m\{\tilde{X}[-k]\} \\ \lvert\tilde{X}[k]\rvert = \lvert\tilde{X}[-k]\rvert \\ \sphericalangle\tilde{X}[k] = -\sphericalangle\tilde{X}[-k] \end{cases}$
16.	$\tilde{x}_e[n] = \frac{1}{2}(\tilde{x}[n] + \tilde{x}[-n])$	$\mathcal{R}e\{\tilde{X}[k]\}$
17.	$\tilde{x}_0[n] = \frac{1}{2}(\tilde{x}[n] - \tilde{x}[-n])$	$j\mathcal{J}m\{\tilde{X}[k]\}$

دوگان

Example 8.4 Periodic Convolution

An illustration of the procedure for forming the periodic convolution of two peri
sequences corresponding to Eq. (8.27) is given in Figure 8.3, where we have illustr

کانولوشن چرخشی

$$\sum_{m=0}^{N-1} \tilde{x}_1[m]\tilde{x}_2[n-m] \overset{DFS}{\longleftrightarrow} \tilde{X}_1[k]\tilde{X}_2[k]$$

ضرب

N پریودیک سیگنال

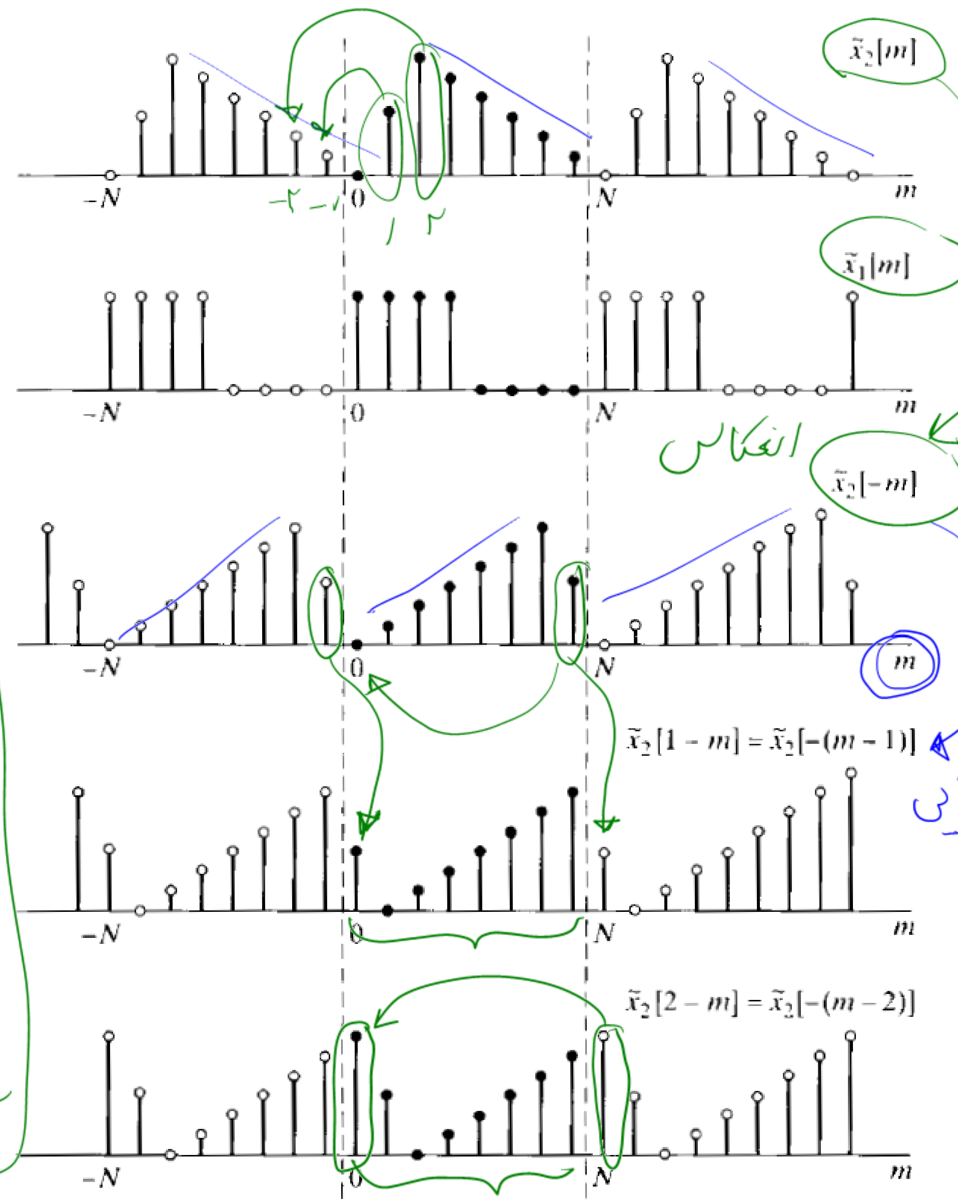
$\tilde{x}_3[n] = \tilde{x}_1[n] \; N \; \tilde{x}_2[n]$

جابجایی چرخشی N-1

n=0

n=1

n=2



جابجایی

$DFS = DTFT \big|_{\omega = k\frac{2\pi}{N}}$ $\quad k = 0, 1, \ldots, N-1$

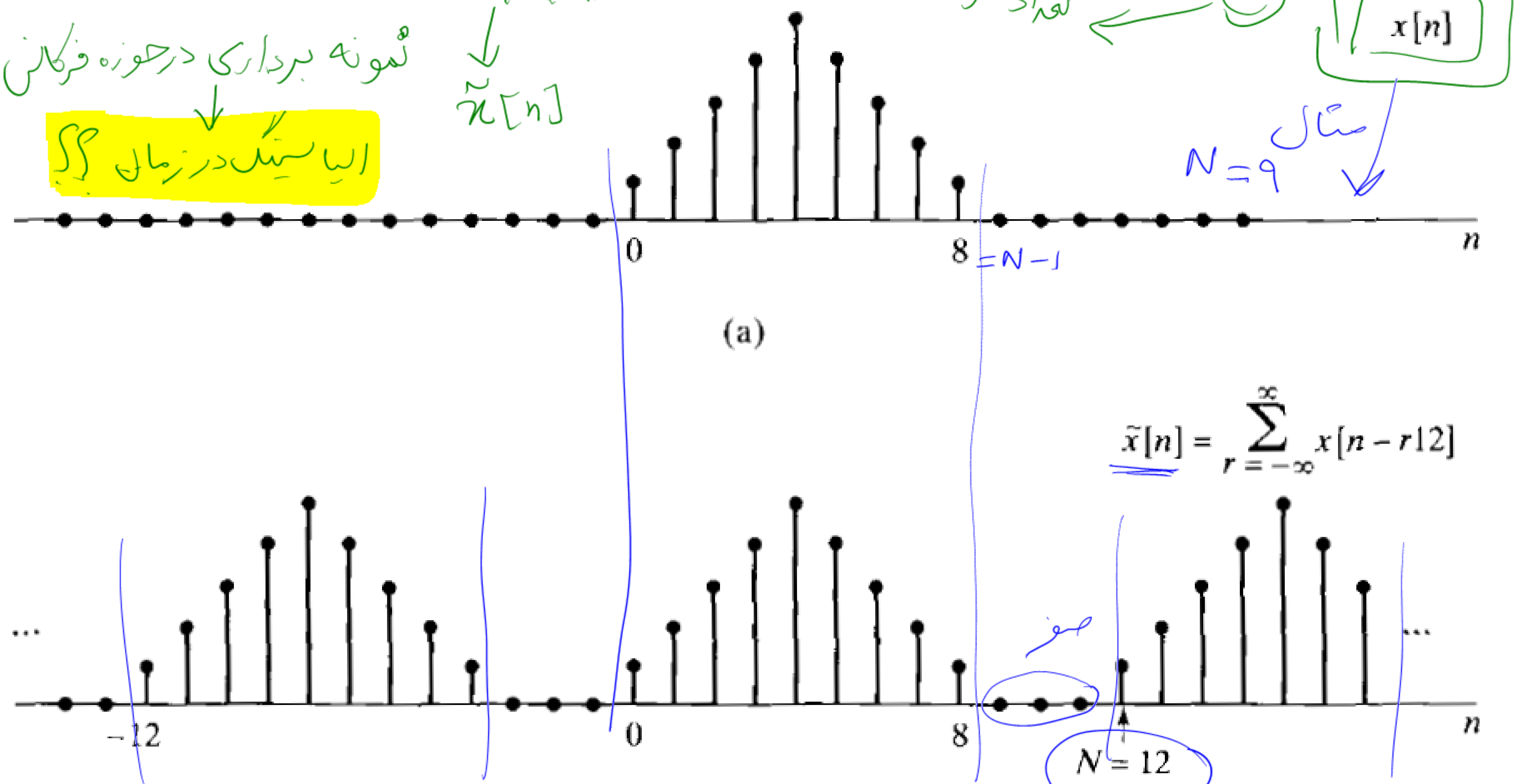
$x[n] \longrightarrow X(e^{j\omega}) \xrightarrow{\omega = k\frac{2\pi}{N}} \tilde{X}[k] \rightarrow \tilde{x}[n]$

تعداد نمونه ها روی دایره واحد

نمونه برداری درحوزه فرکانس

اینا سیگنال درزمان؟؟

مثال $N=9$



تعداد نمونه درحوزه فرکانس

Figure 8.8 (a) Finite-length sequence $x[n]$. (b) Periodic sequence $\tilde{x}[n]$ corresponding to sampling the Fourier transform of $x[n]$ with $N = 12$.

اولیه

الیاسینگ بعد از نمونه برداری

پدیده الیاسینگ در زمان

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - r7]$$

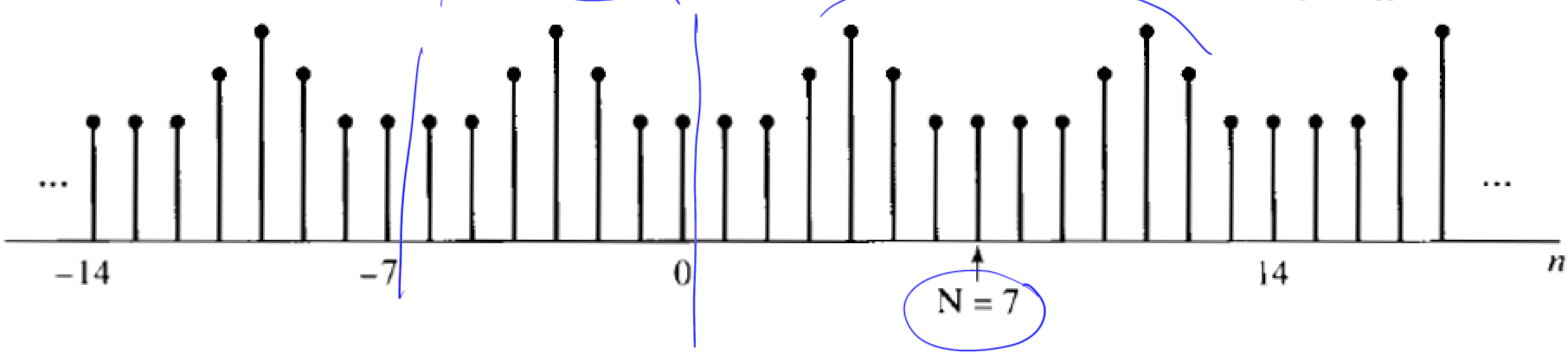


Figure 8.9 Periodic sequence $\tilde{x}[n]$ corresponding to sampling the Fourier transform of $x[n]$ in Figure 8.8(a) with $N = 7$.

حداقل باید N=9 باشد

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

DTFT

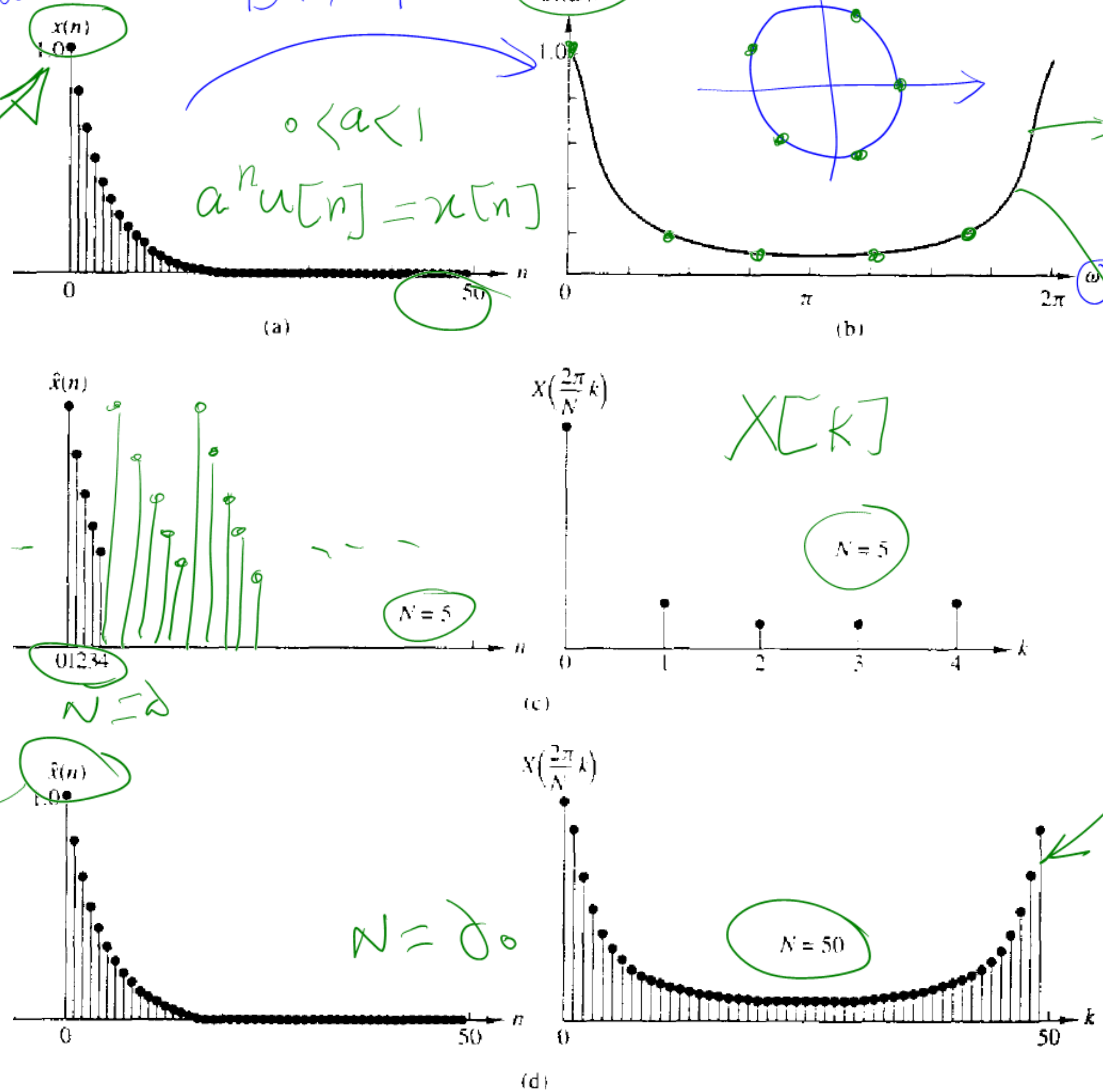
$0<a<1$

$a^n u[n] = x[n]$

$|X(e^{j\omega})|$

الیاسینگ زمان

$X[k]$



$N = ۲$

$N = ۲۰$

Figure 5.4 (a) Plot of sequence $x(n) = (0.8)^n u(n)$; (b) its Fourier transform (magnitude only); (c) effect of aliasing with $N = 5$; (d) reduced effect of aliasing with $N = 50$.

$$\text{DTFT}\Big|_{\omega = k\frac{۲\pi}{N}} = \tilde{X}[k]$$

DFS

محاسبه DFT برای شیفت سیگنال غیر پریودیک $x[n]$

FOURIER REPRESENTATION OF FINITE-DURATION SEQUENCES: THE DISCRETE FOURIER TRANSFORM

$\tilde{x}[n]$ متناوب

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - rN].$$

$r = 0$

متناوب

یک پریود

$$x[n] = \begin{cases} \tilde{x}[n], & 0 \le n \le N-1, \\ 0, & \text{otherwise.} \end{cases}$$

معادل هم هستند

باقی مانده

$m = \frac{n}{N}$ باقی مانده

$0 \le m \le N-1$

$$\tilde{x}[n] = x[(n \text{ modulo } N)].$$

معادل و نمایش دیگر

$$\tilde{x}[n] = x[((n))_N].$$

یک پریود

متناوب

$$X[k] = \begin{cases} \tilde{X}[k], & 0 \le k \le N-1, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\tilde{X}[k] = X[(k \text{ modulo } N)] = X[((k))_N].$$

From Section 8.1, $\tilde{X}[k]$ and $\tilde{x}[n]$ are related by

DFS

سری فوریه

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn},$$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn}.$$

DFT

تجزیه

$$X[k] = \begin{cases} \displaystyle\sum_{n=0}^{N-1} \underbrace{x[n]}_{\text{زمان}} W_N^{kn}, & 0 \le k \le N-1, \\ 0, & \text{otherwise,} \end{cases}$$

فرکانس

جداسازی یک پریود از DFS

ترکیب

$$\underbrace{x[n]}_{\text{زمان}} = \begin{cases} \displaystyle\frac{1}{N} \sum_{k=0}^{N-1} \underbrace{X[k]}_{\text{فرکانس}} W_N^{-kn}, & 0 \le n \le N-1, \\ 0, & \text{otherwise.} \end{cases}$$

Analysis equation: $X[k] = \displaystyle\sum_{n=0}^{N-1} x[n] W_N^{kn}$,

Synthesis equation: $x[n] = \displaystyle\frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$.

سری فوریه ۵ نقطه

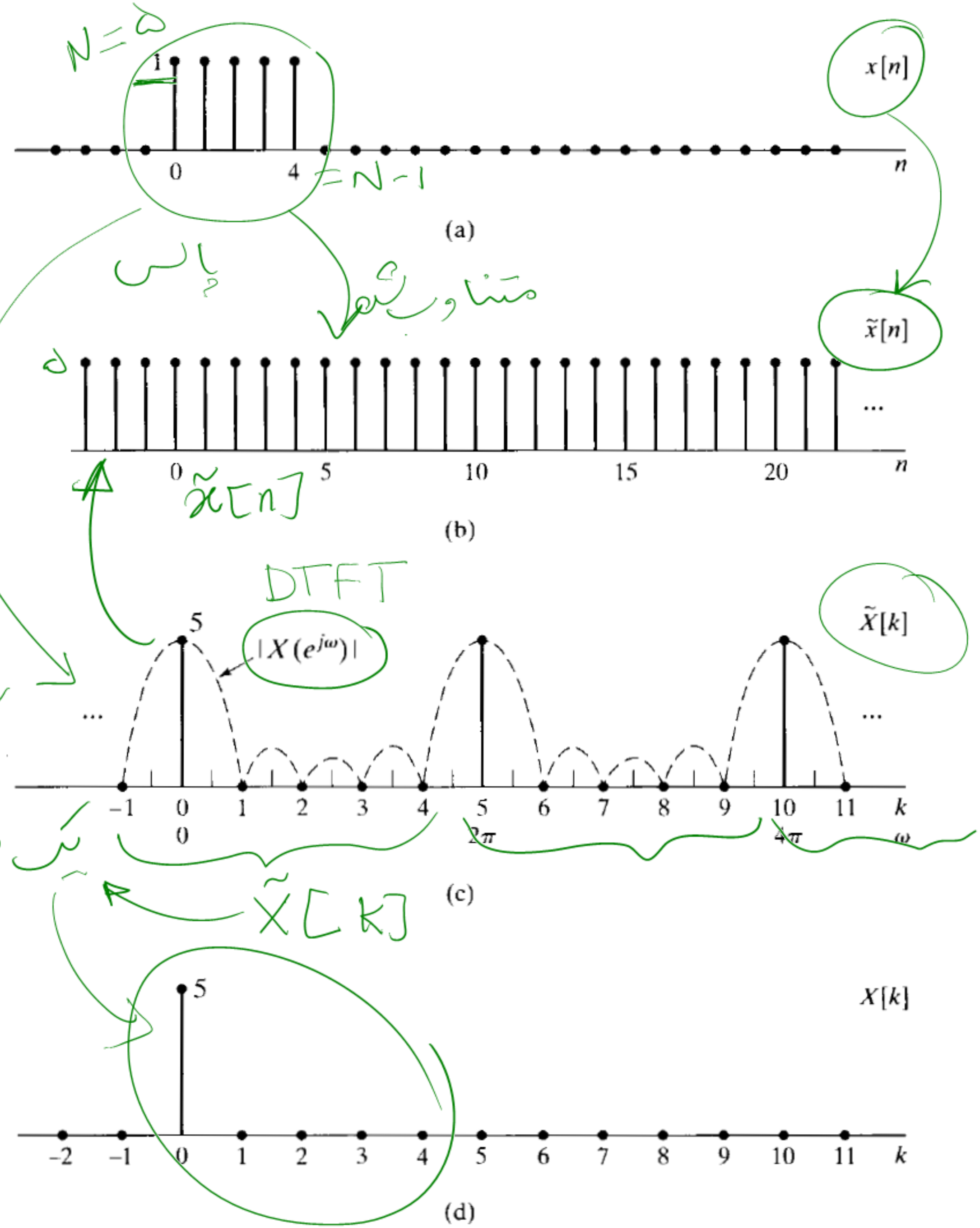
$$\tilde{X}[k] = \sum_{n=0}^{4} e^{-j(2\pi k/5)n} = \frac{1-e^{-j2\pi k}}{1-e^{-j(2\pi k/5)}}$$

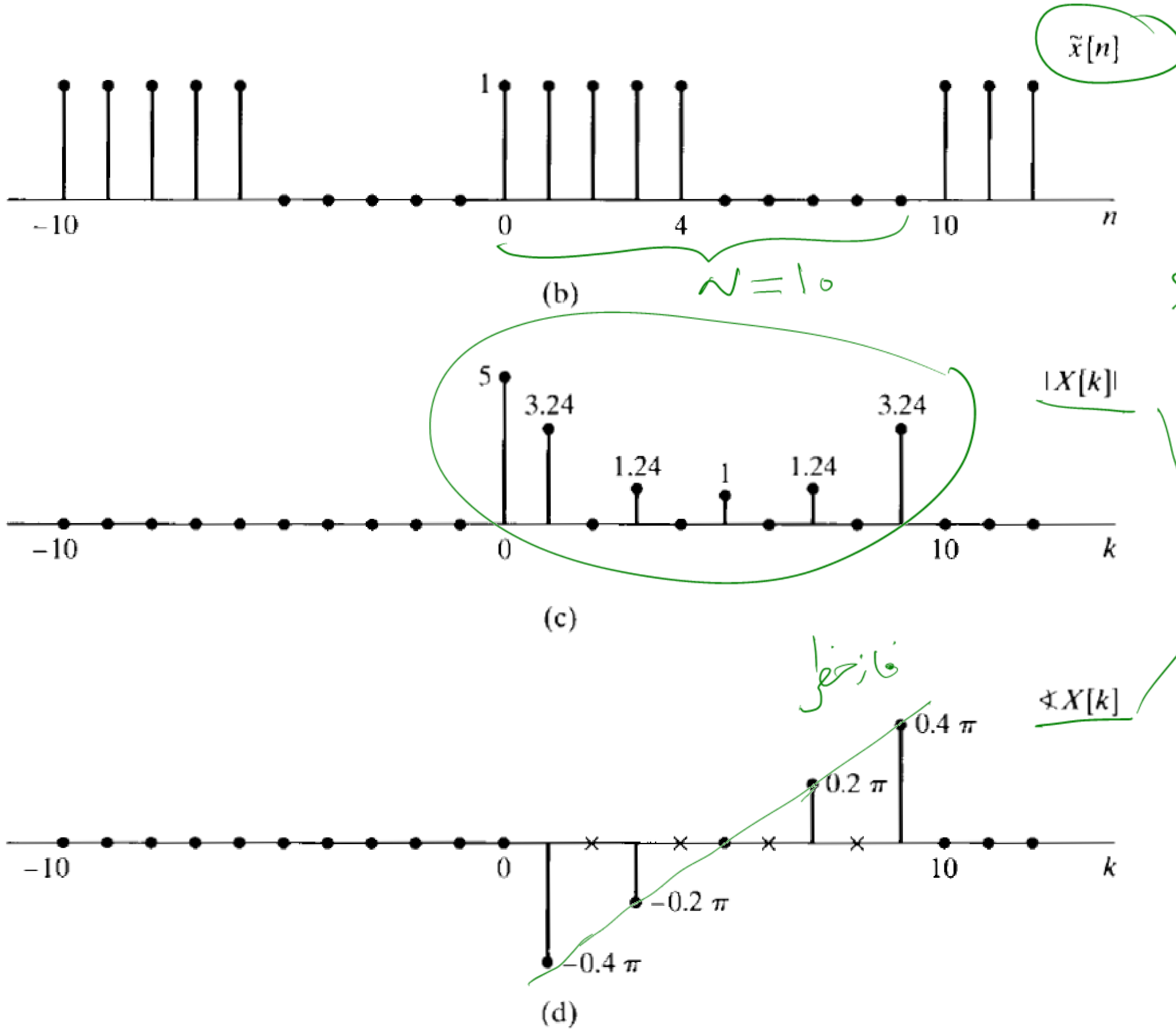
$$= \begin{cases} 5, & k = 0, \pm 5, \pm 10, \ldots, \\ 0, & \text{otherwise;} \end{cases}$$

سینک

یک ضربه

یک پریود





N = 10

تعداد نمونه

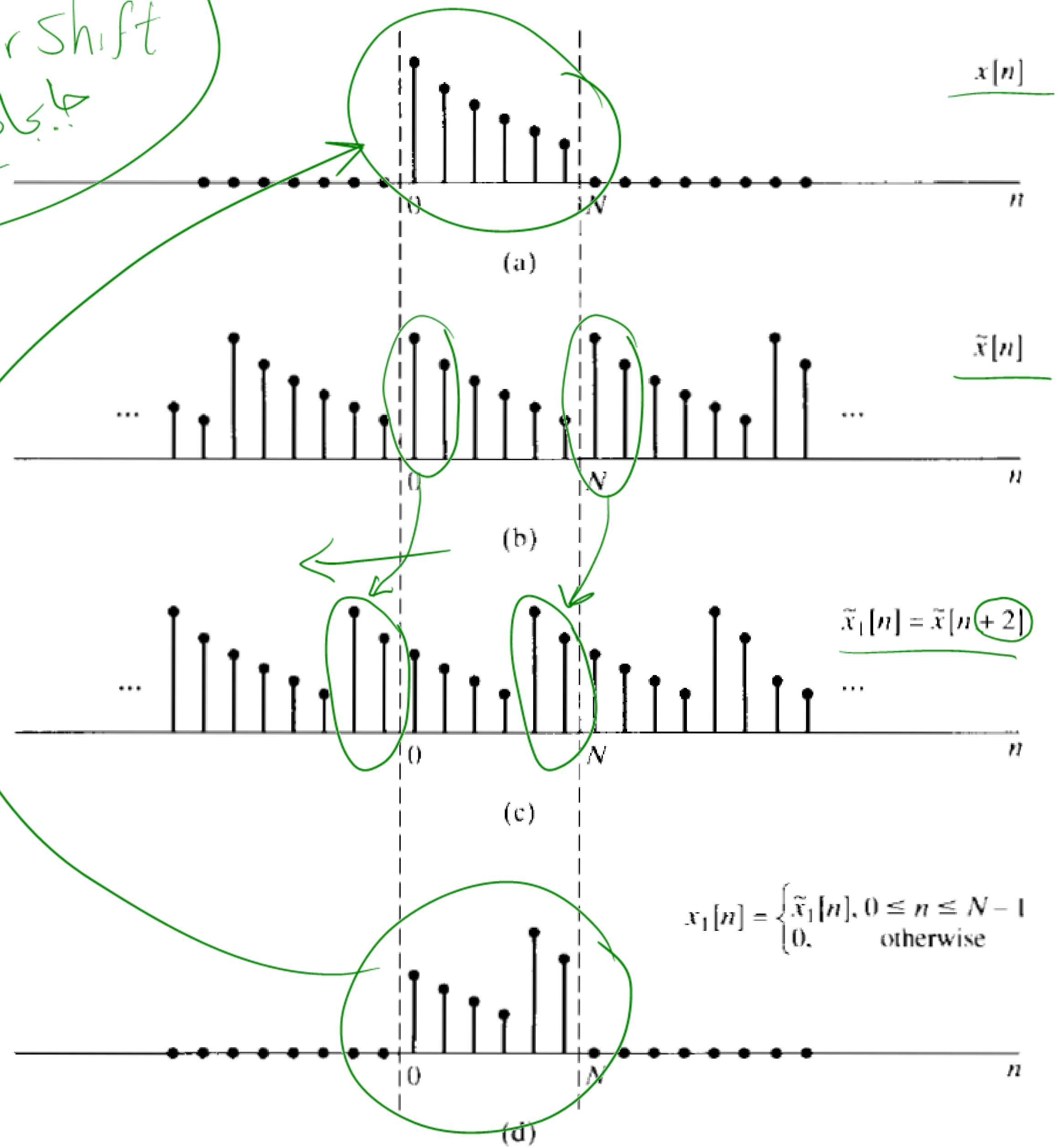
N = 10

اندازه و فاز DFT

فاز خطی

Figure 8.11 Illustration of the DFT. (a) Finite-length sequence $x[n]$. (b) Periodic sequence $\tilde{x}[n]$ formed from $x[n]$ with period $N = 10$. (c) DFT magnitude. (d) DFT phase. (x's indicate indeterminate values.)

Circular Shift
جابجایی چرخشی



متناوب با
پریود N

Figure 8.12 Circular shift of a finite-length sequence; i.e., the effect in the time domain of multiplying the DFT of the sequence by a linear phase factor.

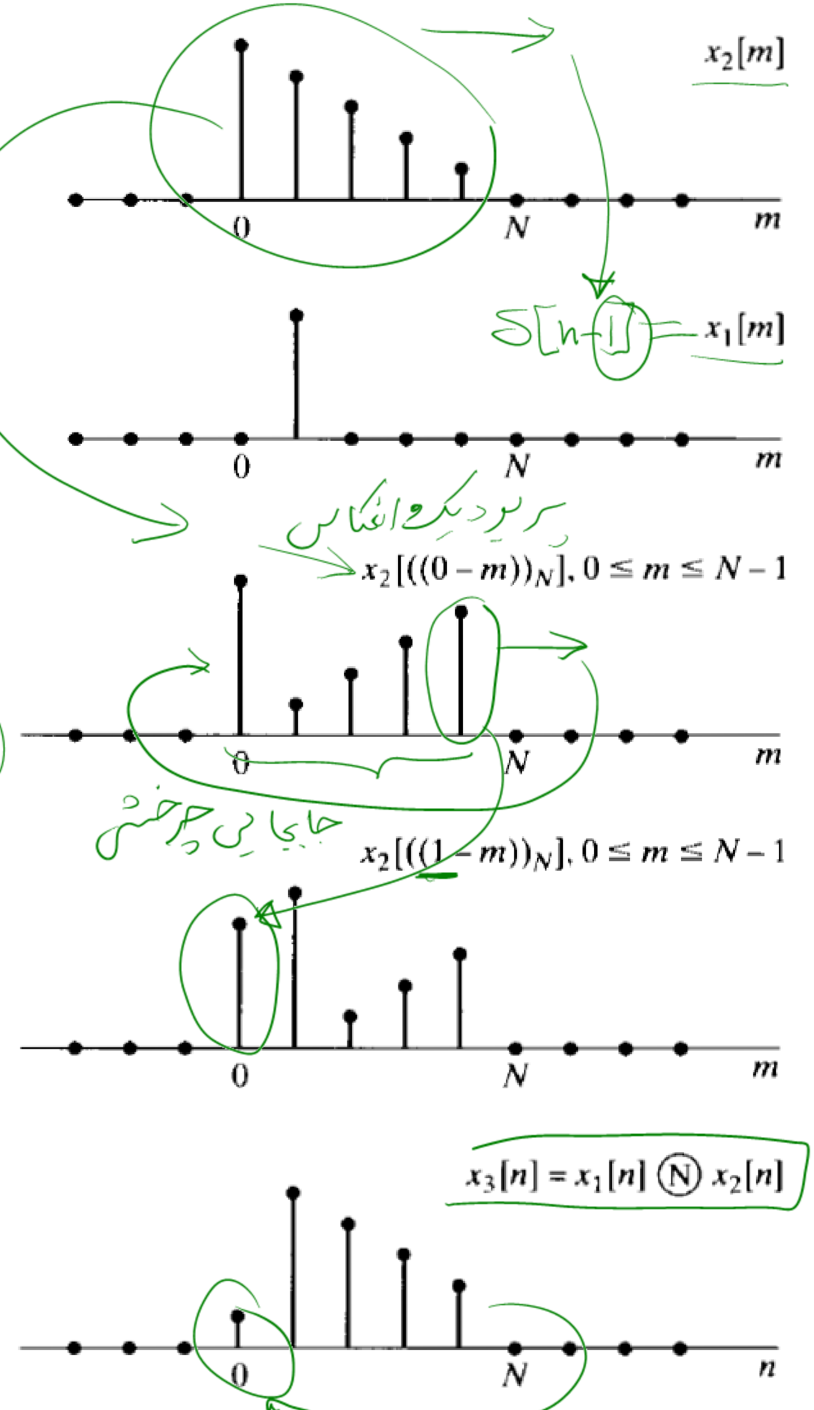
$$x_3[n] = \sum_{m=0}^{N-1} \tilde{x}_1[m]\tilde{x}_2[n-m], \qquad 0 \le n \le N-1,$$

or, equivalently,

$$x_3[n] = \sum_{m=0}^{N-1} x_1[((m))_N]x_2[((n-m))_N], \qquad 0 \le n \le N-1.$$

Since $((m))_N = m$ for $0 \le m \le N-1$, Eq. (8.113) can be written

$$x_3[n] = \sum_{m=0}^{N-1} x_1[m]x_2[((n-m))_N], \qquad 0 \le n \le N-1.$$



$\tilde{x}_1 \textcircled{N} \tilde{x}_2 \longleftrightarrow \tilde{X}_1[k] \cdot \tilde{X}_2[k]$

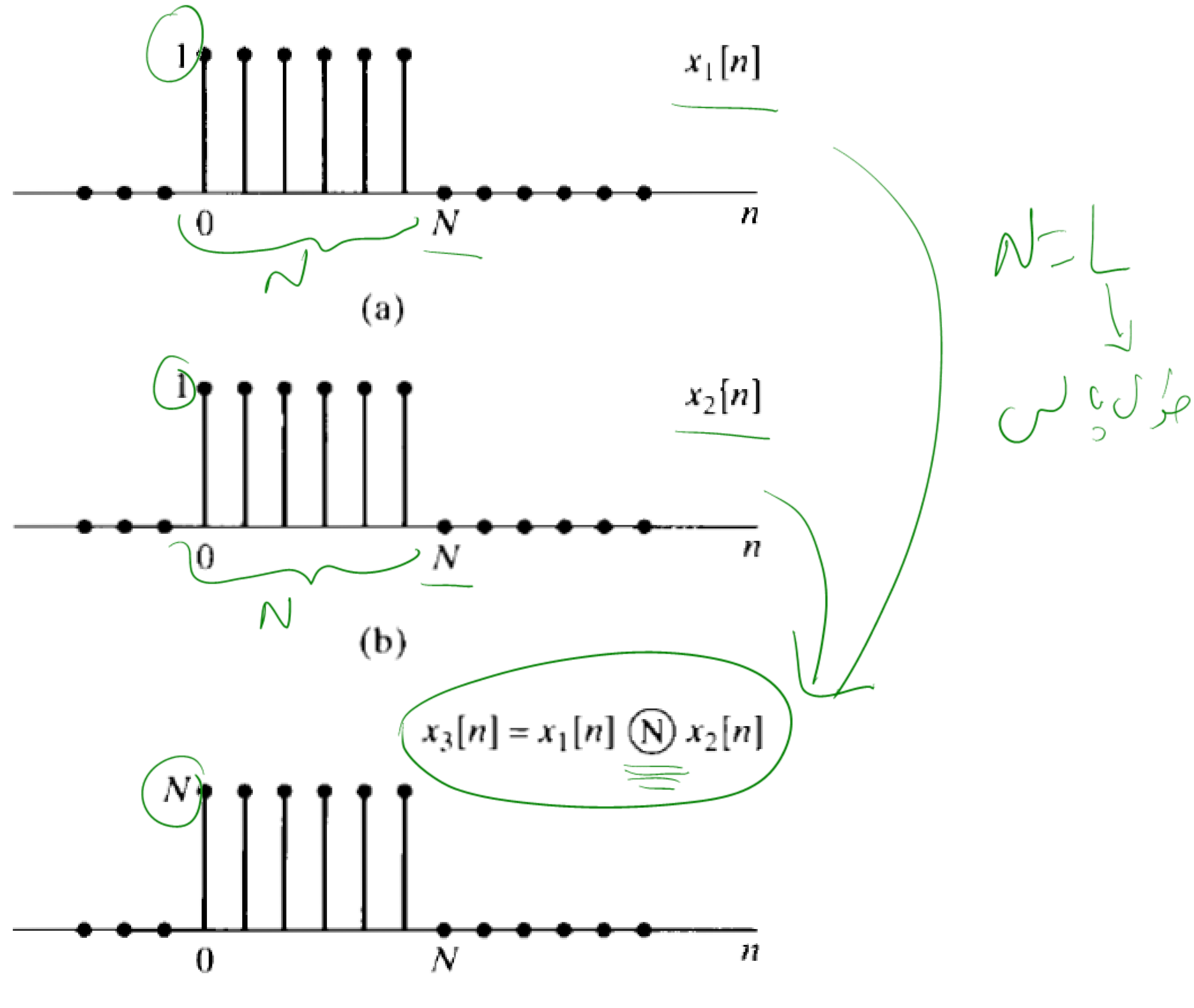
کانولوشن چرخشی

ضرب در حوزه DFS

$x[n]$ و $x_2[n]$ غیرمتناوب

کانولوشن خطی

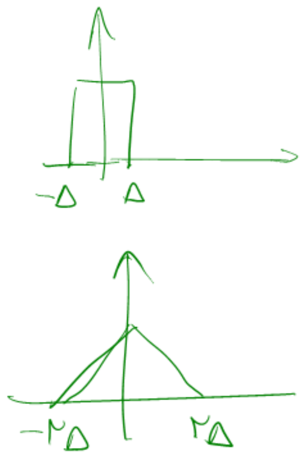
$x_1[n] * x_2[n] = x_3[n]$

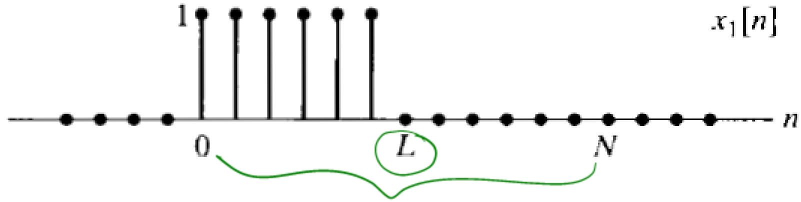


(a)

(b)

(c)

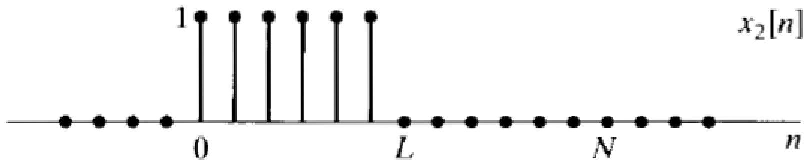




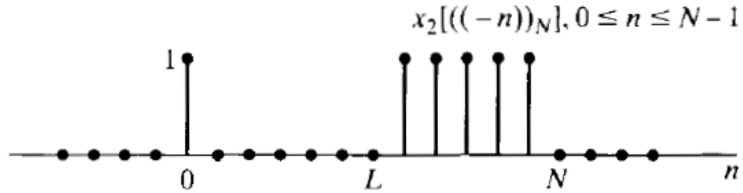
$x_1[n]$
1
0
L
N
n

(a)

$N = rL$

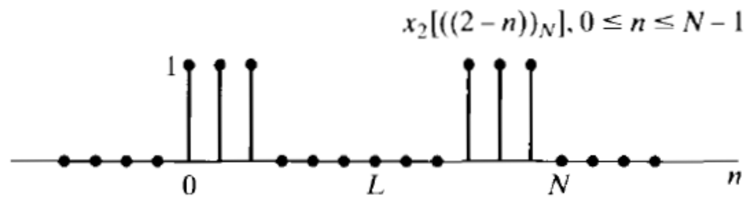


$x_2[n]$
1
0
L
N
n



$x_2[((-n))_N], 0 \le n \le N-1$
1
0
L
N
n

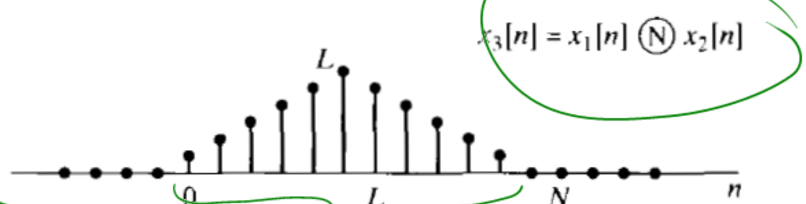
(c)



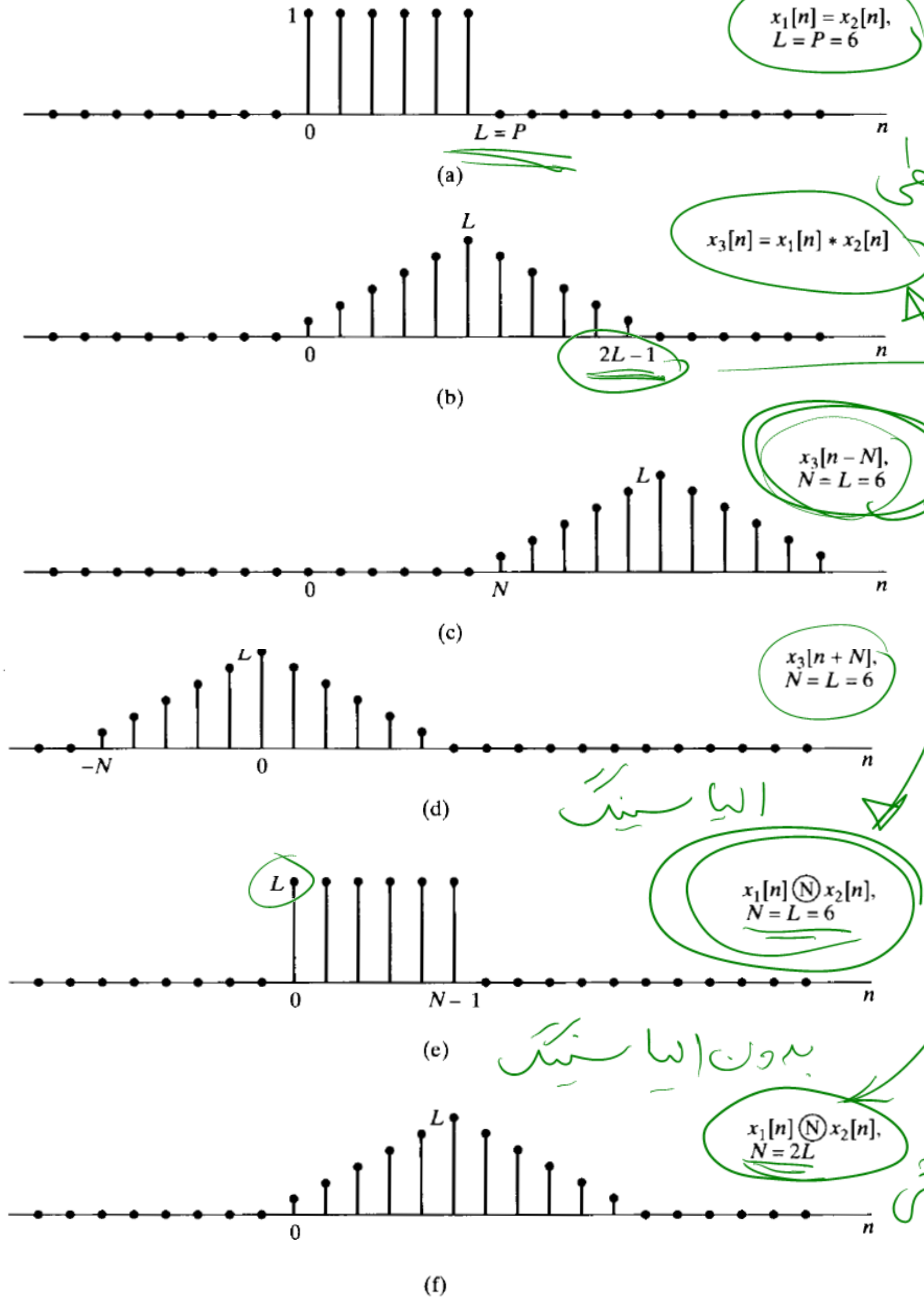
$x_2[((2-n))_N], 0 \le n \le N-1$
1
0
L
N
n

(d)

$rL - 1$



$x_3[n] = x_1[n] \textcircled{N} x_2[n]$
L
0
L
N
n



دوپالس

کانولوشن خطی

الیاسینگ

بدون الیاسینگ

$x_1[n] \circledN x_2[n] = x_1[n] * x_2[n]$

کانولوشن چرخشی

Figure 8.18 Illustration that circular convolution is equivalent to linear convolution followed by aliasing. (a) The sequences $x_1[n]$ and $x_2[n]$ to be

John Proakis

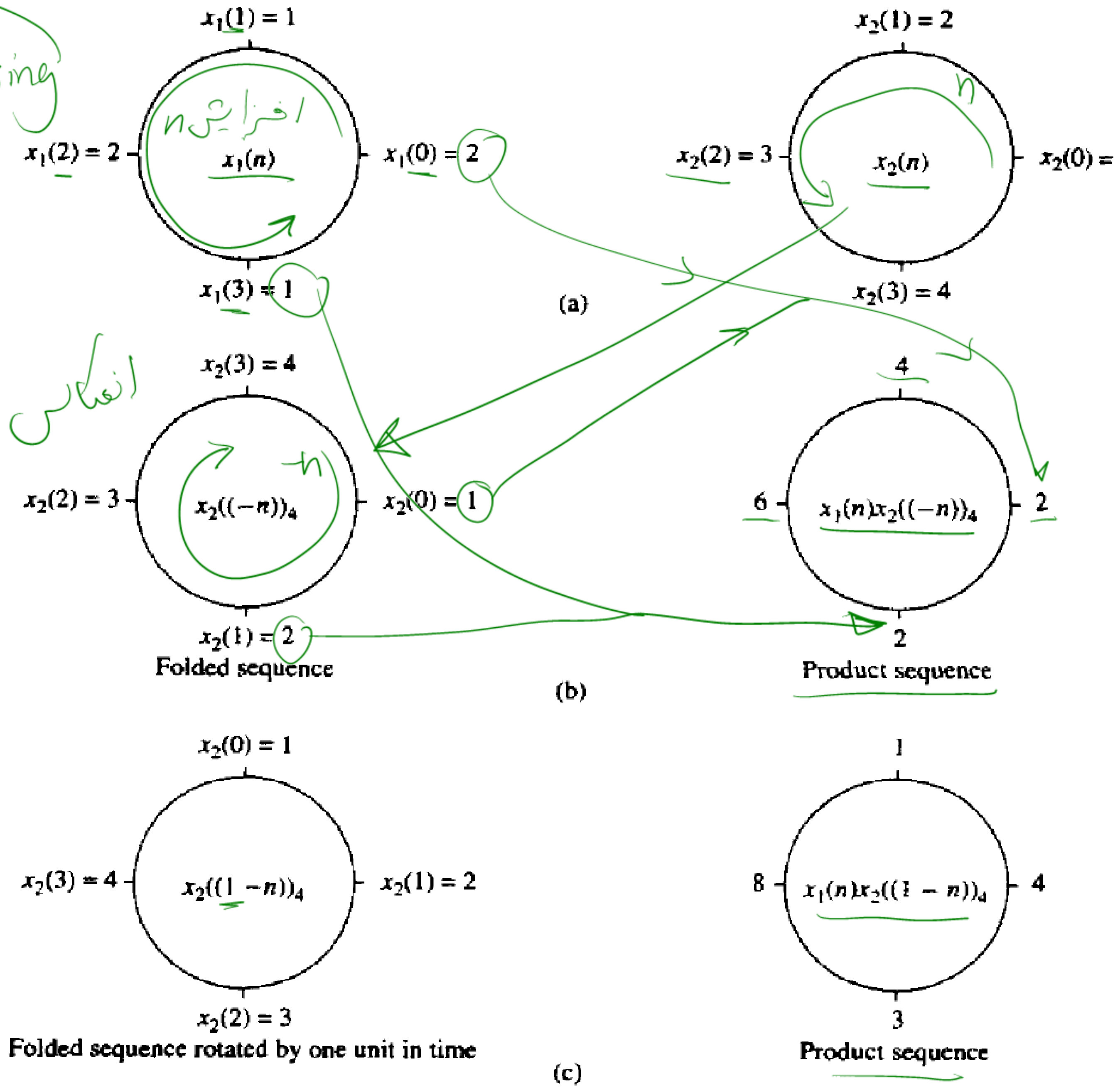
Digital signal Processing

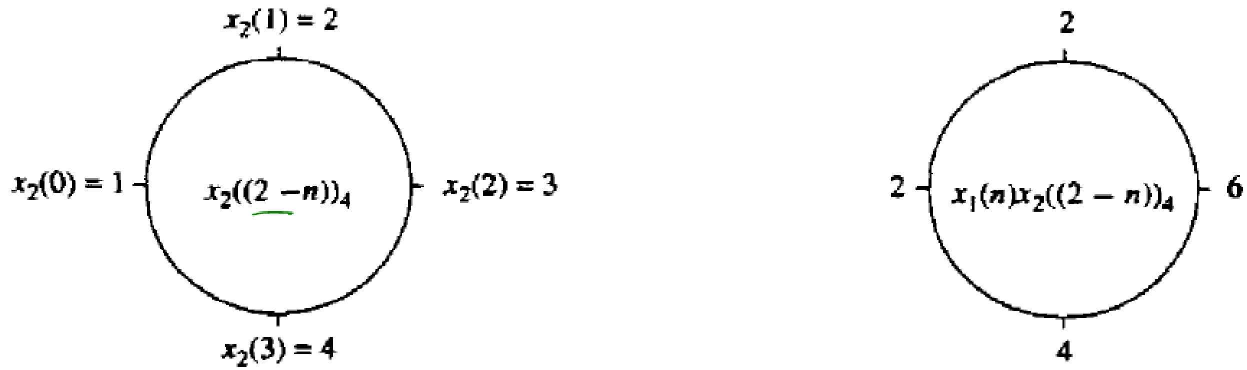
$$x_1(n) = \{\underset{\uparrow}{2}, 1, 2, 1\}$$

n=0 n=1

$$x_2(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$$

n=0

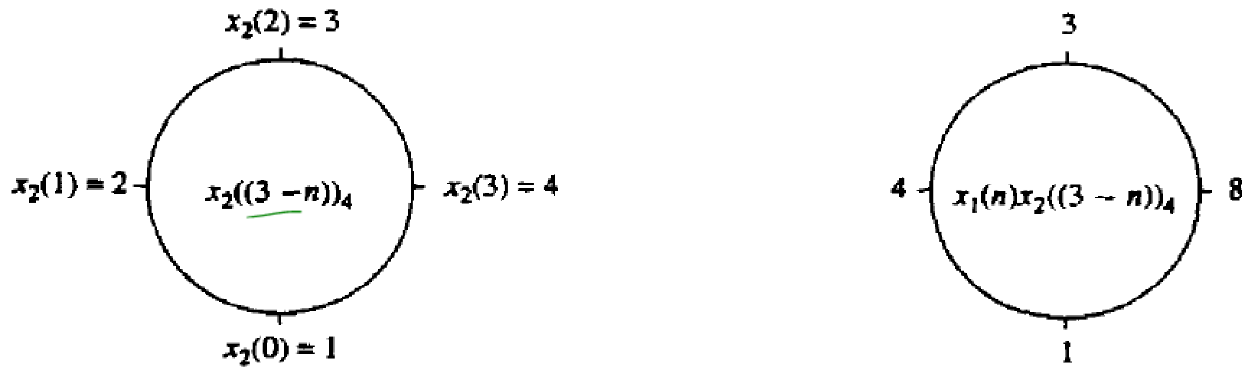




Folded sequence rotated by two units in time

Product sequence

(d)



Folded sequence rotated by three units in time

Product sequence

(e)

$$x_3(n) = \{\underset{\uparrow}{14}, 16, 14, 16\}$$